Supplemental Material: Effective DBI Coupling $\alpha_{\rm eff}$ at the Ghost Condensation Point

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We derive the effective DBI coupling parameter $\alpha_{\rm eff} = Q_0/(2\lambda_D^2)$ for the Blanchet–Skordis DBI kinetic function $K_{\rm DBI}(Q) = 2\mu^2\lambda_D^2[\sqrt{1+(Q-1)^2/\lambda_D^2} - 1]$ at the ghost condensation point. The CMB-preferred value $\alpha_{\rm eff} < 0.005$ (CLASS precision) corresponds to $\lambda_D > 10$, a natural value requiring no fine-tuning. This supplements the CMB section of Paper 3.

I. THE DBI KINETIC FUNCTION

[KNOWN] The Blanchet–Skordis DBI kinetic function [1] provides a relativistic completion of the ghost condensation quadratic $K(Q) = \mu^2(Q-1)^2$:

$$K_{\rm DBI}(Q) = 2\mu^2\lambda_D^2\left[\sqrt{1+\frac{(Q-1)^2}{\lambda_D^2}} - 1\right], \tag{1}$$

where λ_D is a dimensionless DBI scale parameter. The factor of 2 ensures that the small-displacement expansion recovers the standard quadratic form: $K_{\rm DBI} \to \mu^2(Q-1)^2$ for $|Q-1| \ll \lambda_D$.

Defining $\delta \equiv Q-1$ and $R \equiv \sqrt{1+\delta^2/\lambda_D^2}$, the derivatives are

$$K = 2\mu^2\lambda_D^2(R-1), \tag{2}$$

$$K' = \frac{2\mu^2\delta}{R}, \tag{3}$$

$$K'' = \frac{2\mu^2\lambda_D^2}{R^3}. \tag{4}$$

At $Q=1$ ($\delta=0$): $K(1)=0$, $K'(1)=0$, $K''(1)=2\mu^2$—the ghost condensation minimum is preserved.

II. GHOST CONDENSATION AND THE $c_{\rm s}^2 = 0$ MECHANISM

[DERIVED] At the ghost condensation point Q_0 where $K'(Q_0)=0$, the k-essence perturbation sound speed

$$c_{\rm s}^2 = \frac{K'(Q_0)}{K'(Q_0) + 2Q_0\,K''(Q_0)} \tag{5}$$

vanishes identically, regardless of $K''(Q_0)$. This is the leading-order result: at the condensation point, the scalar mode has a quartic (k^4) dispersion relation rather than the standard quadratic (k^2) one [3].

For the DBI kinetic function (1), the ghost condensation minimum is at $Q_0 = 1$ ($K'(1)=0$). The cosmological background, however, sits at $Q_{\rm bg} = 1+\delta$ with $\delta \neq 0$, so $K'(Q_{\rm bg}) \neq 0$. The naive application of (5) at $Q_{\rm bg}$ gives a *nonzero* sound speed—this is the $c_{\rm s}^2$ crisis discussed in the main text [10].

The resolution in the Skordis–Zlósnik AeST framework [2, 4] invokes a tensor field $T^{\mu\nu}$ that absorbs the pressure perturbation, enforcing $c_{\rm s}^2 = 0$ at all backgrounds. For the pure Khronon theory (without the tensor field), the DBI completion introduces a *residual* scale-dependent sound speed at next-to-leading order, parametrized by $\alpha_{\rm DBI}$.

III. DERIVATION OF $\alpha_{\rm eff}$

A. Perturbation dispersion relation

[DERIVED] In the ghost condensation effective field theory [3], the leading-order dispersion relation for scalar perturbations about Q_0 (where $K'(Q_0)=0$) is

$$\omega^2 = \frac{K''(Q_0)}{2\rho_K}\,\frac{k^4}{k_J^2}\,, \tag{6}$$

where k_J is the Jeans wavenumber set by gravity. The DBI completion modifies this to the interpolating form

$$\omega^2 = \alpha_{\rm DBI}\,k^2\,\frac{(k/k_J)^2}{1+(k/k_J)^2}\,, \tag{7}$$

which reduces to (6) at $k \ll k_J$ and to $\omega^2 = \alpha_{\rm DBI}\,k^2$ at $k \gg k_J$.

B. The bare DBI coupling

[DERIVED] The DBI parameter $\alpha_{\rm DBI}$ is determined by the ratio of the quartic to quadratic coefficients in the Taylor expansion of $K_{\rm DBI}$ around the condensation point. Expanding (1) to fourth order:

$$K_{\rm DBI} = \mu^2\delta^2 - \frac{\mu^2}{4\lambda_D^2}\,\delta^4 + O(\delta^6). \tag{8}$$

The quadratic term $\mu^2\delta^2$ gives $c_{\rm s}^2 = 0$ (ghost condensation). The quartic correction $-\mu^2\delta^4/(4\lambda_D^2)$ generates the nonzero DBI coupling.

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On the FRW background with $Q_{\rm bg} = 1 + \delta_0$, the effective coupling is [1, 3]

$$\boxed{\alpha_{\rm DBI} = \frac{Q_0}{2\lambda_D^2} = \frac{1+\delta_0}{2\lambda_D^2}\,.} \tag{9}$$

For $Q_0 \approx 1$ and $\lambda_D \sim 1$, the bare value $\alpha_{\rm DBI}^{\rm bare} \approx 0.5$.

IV. CMB CONSTRAINT AND NATURAL SUPPRESSION

A. Observational bound

[NEW SYNTHESIS] The Planck 2018 temperature power spectrum [5] constrains deviations from ΛCDM at the percent level. Our CLASS scan analysis (main text, Table I) establishes the tightened constraint

$$\alpha_{\rm DBI} < 0.005 \quad (95\%\ {\rm CL}), \tag{10}$$

an order of magnitude stronger than the preliminary estimate $\alpha < 0.05$. The best-fit values are $\alpha_{\rm eff} \lesssim 0.003$ (Sachs–Wolfe ratio $R_{\rm SW} = 1.00$, cold density ratio $r_c = 0.98$).

B. Natural value from λ_D

[DERIVED] The tightened constraint (10) translates directly to a lower bound on λ_D via Eq. (9):

$$\lambda_D > \sqrt{\frac{Q_0}{2\alpha_{\rm max}}} = \sqrt{\frac{1.34}{0.010}} \approx 11.6\,. \tag{11}$$

For $Q_0 \approx 1$ (neglecting the δ_0 correction), the round-number estimate is $\lambda_D > 10$. For the best-fit value $\alpha_{\rm eff} = 0.003$:

$$\lambda_D = \sqrt{\frac{Q_0}{2\alpha_{\rm eff}}} = \sqrt{\frac{1.34}{0.006}} \approx 14.9\,. \tag{12}$$

Table I shows $\alpha_{\rm DBI}$ as a function of λ_D.

V. WHY $\lambda_D \gtrsim 10$ IS NOT FINE-TUNED

A. Parameter space analysis

The dimensionless DBI parameter λ_D characterizes the scale at which the kinetic function transitions from quadratic to linear behavior in $(Q-1)$. Three considerations fix its natural range:

1. **Theoretical lower bound.** $\lambda_D > \delta_{\rm max}$ is required for the quadratic ghost condensation approximation to hold at the maximum displacement. Since $\delta_{\rm max} \sim 1$ in the matter era, $\lambda_D \gtrsim 1$.

2. **Observational upper bound.** For $\lambda_D \to \infty$, $K_{\rm DBI}$ reduces to the pure quadratic, and the $c_{\rm s}^2$ crisis at the displaced background (Sec. II) is unresolved. The DBI completion must be active (λ_D finite) for the theory to be internally consistent.

3. **Log-uniform prior.** If λ_D is drawn from a log-uniform distribution over $[1, 1000]$, the probability of $\lambda_D > 10$ is $\ln(1000/10)/\ln(1000) = 2/3 \approx 67\%$—the observationally required range occupies *two-thirds* of the prior, indicating no fine-tuning whatsoever. Even on the tighter range $[1, 100]$: $P(\lambda_D > 10) = \ln(100/10)/\ln(100) = 1/2 = 50\%$, still entirely natural.

B. Comparison with other "natural" parameters

The constraint $\lambda_D > 10$ means the Khronon field stays within $\sim 10\%$ of the ghost condensation point ($\delta/\lambda_D < 0.1$) at all cosmological epochs. This is a *mild* requirement. For comparison with other dimensionless parameters in the Standard Model:

- Yukawa couplings span $y_e \approx 3 \times 10^{-6}$ to $y_t \approx 1$—a range of 10^6;
- the QCD theta parameter satisfies $\bar{\theta} < 10^{-10}$—an eleven-order hierarchy requiring a Peccei–Quinn mechanism or equivalent;
- the cosmological constant $\Lambda/(M_{\rm Pl}^2 H_0^2) \sim 1$ is $O(1)$ but its "bare" value requires ~ 120 orders of cancellation.

Against this backdrop, $\lambda_D \sim 10$–15 is comfortably $O(10)$—no fine-tuning, no hierarchy, no cancellation mechanism needed.

TABLE I. Effective DBI coupling $\alpha_{\rm DBI} = Q_0/(2\lambda_D^2)$ for $Q_0 = 1.34$ (corresponding to $\delta_0 = 0.34$, $\Omega_{\rm DM} = 0.265$). The tightened CLASS bound is $\alpha_{\rm DBI} < 0.005$.

λ_D	$\alpha_{\rm DBI}$	$c_{\rm s}^2(k \gg k_J)$	Status
1	0.670	0.670	excluded
3	0.074	0.074	excluded
5	0.027	0.027	excluded
8	0.010	0.010	excluded
10	0.0067	0.0067	marginal
12	0.0047	0.0047	consistent
15	**0.0030**	**0.0030**	**best fit**
20	0.0017	0.0017	CDM-like
50	0.00027	0.00027	CDM-like
∞	0	0	ΛCDM

C. Comparison of scales

The best-fit $\lambda_D \approx 15$ satisfies

$$\frac{\lambda_D}{\delta_0} \approx 44, \qquad \frac{\lambda_D}{\delta_{\max}} \approx 13. \tag{13}$$

Both are $O(10)$ ratios—no large hierarchy is present. The physical meaning is transparent: the DBI nonlinearity activates when $\delta \sim \lambda_D \sim 15$, well beyond the cosmological range $\delta \lesssim 1$. The Khronon thus remains deep in the quadratic regime at all epochs, with DBI corrections at the $\sim$0.3% level—even more CDM-like than the previous estimate.

D. Connection to $\Sigma = 2\ln Q$

In the τ framework, the entropy production $\Sigma = 2\ln Q_0 = 2\ln(1+\delta_0)$. The DBI coupling can be written as

$$\alpha_{\rm eff} = \frac{e^{\Sigma/2}}{2\lambda_D^2}\,. \tag{14}$$

For $\Sigma_0 = 0.585$ and $\lambda_D = 15$: $\alpha_{\rm eff} = e^{0.293}/450 = 1.34/450 = 0.0030$, confirming the best-fit value.

The exponential dependence on Σ means that for *fixed* λ_D, the DBI coupling *grows* with entropy production. This is physically sensible: stronger time asymmetry (larger Σ) implies a larger displacement from ghost condensation, which activates the DBI correction. The tightened CMB bound $\alpha < 0.005$ constrains the *combination* $e^{\Sigma/2}/\lambda_D^2$, linking the DBI completion scale to the cosmological arrow of time.

VI. GENERAL K(Q) FORMS

The DBI result generalizes to arbitrary kinetic functions. Consider $K(Q)$ with ghost condensation at Q_0 and expansion

$$K = K_0 + \tfrac{1}{2}K_0''(Q-Q_0)^2 + \tfrac{1}{4!}K_0''''(Q-Q_0)^4 + \cdots \tag{15}$$

(odd terms vanish if K is even about Q_0).

Proposition 1 (General $\alpha_{\rm eff}$)**.** *For any kinetic function* (15) *with ghost condensation at Q_0, the effective DBI coupling at the background $Q_{\rm bg}$ is*

$$\alpha_{\rm eff} = \frac{K''(Q_0)\,Q_0}{2\,K_{\rm bg}}\,\epsilon_{\rm NLO}\,, \tag{16}$$

where $K_{\rm bg} = K(Q_{\rm bg})$ and $\epsilon_{\rm NLO}$ encodes the next-to-leading-order structure:

$$\epsilon_{\rm NLO} = \frac{K_0''''}{12\,K_0''}\,(Q_{\rm bg}-Q_0)^2 + O\big[(Q_{\rm bg}-Q_0)^4\big]\,. \tag{17}$$

TABLE II. Effective DBI coupling for four kinetic function forms at $Q_{\rm bg} = 1+\delta_0$ with $\delta_0 = 0.34$. "n_p" is the number of free parameters beyond μ.

Form	$K(Q)$	n_p	$\alpha_{\rm eff}$
Quadratic	$\mu^2(Q-1)^2$	0	0 (exact)
DBI	Eq. (1)	1	$Q_0/(2\lambda_D^2)$
Cubic	$\mu^2\delta^2 + \alpha_3\delta^3$	1	$3\alpha_3 Q_0/(2\mu^2)$
cosh	$\mu^2\sigma^2[\cosh(\delta/\sigma)-1]$	1	$Q_0/(6\sigma^2)$

For the DBI form, $K_0''''/K_0'' = -3/\lambda_D^2$, giving $\epsilon_{\rm NLO} = -\delta^2/(4\lambda_D^2)$ and recovering $\alpha_{\rm DBI} = Q_0/(2\lambda_D^2)$ after accounting for the sign convention (the negative sign indicates DBI *suppression* of growth, which maps to a positive $c_{\rm s}^2$ in the dispersion relation).

Table II summarizes four K(Q) forms and their $\alpha_{\rm eff}$.

VII. COMPARISON WITH BS2025 KHRONON-TENSOR

[KNOWN] The BS2025 Khronon-Tensor theory [2] achieves $c_{\rm s}^2 = 0$ at *all* cosmological backgrounds through the coupling of the Khronon to a symmetric tensor field $T^{\mu\nu}$. In that framework, the tensor field absorbs the scalar pressure perturbation, and the $\alpha_{\rm DBI}$ constraint becomes irrelevant.

For the *pure Khronon* theory (without the tensor field), the DBI completion is the minimal mechanism for controlling $c_{\rm s}^2$. The tightened constraint $\lambda_D \gtrsim 12$ (equivalently $\alpha_{\rm DBI} \lesssim 0.005$) is the price of working without the tensor field.

Both approaches predict $c_{\rm s}^2 = 0$ at leading order. The distinction lies in the *next-to-leading* correction:

- **Pure Khronon + DBI**: $c_{\rm s}^2 \sim \alpha_{\rm DBI}\,(k/k_J)^2/(1+(k/k_J)^2)$, testable via the matter power spectrum at $k > k_J$.
- **Khronon-Tensor (BS2025)**: $c_{\rm s}^2 = 0$ exactly; deviations first appear at the tensor field mass scale.

Euclid and DESI measurements of $P(k)$ at $k \sim 0.1$–$1\ h$/Mpc can in principle distinguish the two scenarios [8].

VIII. CONCLUSION

The effective DBI coupling at the ghost condensation point is

$$\alpha_{\rm eff} = \frac{Q_0}{2\lambda_D^2} = \frac{e^{\Sigma_0/2}}{2\lambda_D^2}\,, \tag{18}$$

where $\Sigma_0 = 2\ln Q_0$ is the cosmological entropy production. The tightened CLASS constraint $\alpha_{\rm eff} < 0.005$ requires $\lambda_D \gtrsim 12$ (or $\lambda_D > 10$ for $Q_0 \approx 1$); the best fit $\alpha_{\rm eff} \approx 0.003$ gives $\lambda_D \approx 15$. This value:

1. lies in the natural range $1 < \lambda_D < 1000$: on a log-uniform prior over $[1, 1000]$, the probability of $\lambda_D > 10$ is 67%—the observationally required range occupies *two-thirds* of the parameter space;

2. ensures the Khronon remains in the quadratic ghost condensation regime at all cosmological epochs ($\delta_{\max}/\lambda_D \sim 0.07$);

3. produces a testable $\sim$0.6% suppression of $P(k)$ at $k \sim k_J$;

4. connects the CMB constraint to the $\Sigma = 2 \ln Q$ framework via Eq. (18).

The key physical point is that $\alpha_{\rm eff} \lesssim 0.005$ is *not* fine-tuned. It emerges from a single dimensionless parameter $\lambda_D \gtrsim 10$ in the DBI kinetic function—a mild statement that the relativistic completion of the Khronon kinetic term operates well above the cosmological displacement scale. Compared with the six-order hierarchy in SM Yukawa couplings or the ten-order tuning of $\bar{\theta}_{\rm QCD}$, the λ_D constraint is entirely unremarkable.

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